

RMB Appreciation: A Quantified Macro Assessment

RMB-Macro-Sim · Monte Carlo scenario simulation × real-data calibration · 6-module transmission model · Capital-inflow module

Research framework · Not investment advice

Macro Research | RMB Asset Revaluation Special | 2026-09-06 | Sources: FRED, Tencent Finance, akshare (northbound) | Report ID: RMBMS-2026-0906-EN

Executive summary

Net benefit (5y, 6% base) +2.68 T\$	Capital inflow (financing) +2.17 T\$	Capital deepening +0.82 T\$	Export profit loss -0.31 T\$	Benefit / cost 8.6x
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One-line thesis: under a controlled 6% annual RMB appreciation, the 5-year net socio-economic benefit is +2.68 T\$ (about 8.6 times the export-sector cost). Losses in low-end manufacturing are more than offset by financing and capital-deepening gains as global capital migrates into RMB assets. The result is robust to pace: even at the realized 2.7%-per-year inertia path the net benefit stays +2.63 T\$.

- Aggregate:** not a Plaza-style deindustrialization replay. Once dollar-scarcity inflow enters the revenue side, net effects turn positive at meaningful scale.
- Structure:** low-end manufacturing takes about a -28.1% five-year profit shock; semiconductors and hard assets benefit (+8.7%, +15.2%). Appreciation is a profit reallocation plus balance-sheet revaluation.
- Calibration:** USD/CNY is already 6.71 (a 7.20 base would be ~7% stale). Realized 1y drift is about -6.2% per year - the "6% scenario" is current policy fact, not fiction.
- Assets:** A-shares +8.1% annualized; CNY gold -5.0% (USD gold stays positive); metals +2.0%; CN 10Y bonds -0.7%.

1. Model and method

Transmission chain: FX path, import inflation, export competitiveness, industry profits, then asset prices plus capital inflows. Each stage is an independent module; all parameters live in a single config.yaml. Outputs are 5-year windows from 2,000 Monte Carlo paths (median with 25-75% bands).

Module	Algorithm	Output
ExchangeRate	GBM + appreciation drift	USD/CNY paths
Inflation	PPI pass-through + CPI lag	Import-inflation path
Export	Marshall-Lerner	Export volume change

Module	Algorithm	Output
IndustryProfit	FX x sensitivity + CPI + exports + rates	11-sector profit shocks
AssetPrice	Weighted equity + duration + factors	A-share/bond/gold/commodity
CapitalInflow	Stock x attractiveness x dollar-drain	3-channel inflows + net waterfall

v2.0 adds the revenue engine: with global dollar liquidity scarce (2026-2027 centripetal window), RMB assets become the preferred container. Inflows deliver financing, capital deepening and revaluation. Net benefit = inflow + deepening + export loss (negative).

2. Real-data calibration (Sep 2026)

2.1 FX facts (FRED DEXCHUS, daily)

Window	Annual vol	Annual drift	Obs
1y window	2.04%	-6.19%	251
3y window	2.84%	-2.70%	751
5y window	4.00%	0.78%	1250
10y window	4.03%	0.07%	2496

Vol = annualized std of daily log returns; negative drift = CNY appreciation. Spot (Tencent, 2026-09-06): **6.7108**.

2.2 Capital inflow (northbound, valid through 2024-08)

Northbound annual net inflow averaged 0.19 T RMB with holdings of 1.71 T RMB. Scaled by 1.7 to full foreign scope (northbound about 60%), the foreign A-share base is about 0.43 T\$ (model base 0.55 in the same ballpark). Historical normal inflow rate: 6.5% per year; the 11.1% base is a scenario value for appreciation plus dollar drain, stress-tested below.

2.3 Export elasticity (1993-2026, 402 twelve-month diffs)

USD-value export elasticity to FX is -0.06 (t=-0.6, R2=0.00) - statistically insignificant, consistent with USD invoicing and demand-driven exports. The model keeps a volume elasticity of -0.45 (value mixes price effects; the volume channel is not falsified).

2.4 Industry FX betas (2019-2026 monthly, ETF proxies, market-adjusted)

Industry (proxy)	Beta per +1% dep	t-stat
High-tech exports	0.95	+1.15
Semiconductor	0.85	+0.91
Metals	-0.64	-1.00
Banks/Insurance	-0.78	-2.29
Gold	0.11	+0.28

Beta = excess monthly return per +1% USD/CNY (CNY depreciation). Directional support for model priors: banks (significant), metals, export chains; semiconductor is opposite-signed (domestic-substitution era) - advisory only.

3. Scenario results - pace and capital sensitivity



Scenario	USD/CNY end	5y apprec	Low-end mfg	Net benefit
A Original (7.20 / 6% / vol4%)	5.312	35.5%	-28.1%	+2.68 T\$
B Real-calibrated (6.71 / 6% / vol2.8%)	4.961	35.3%	-27.9%	+2.68 T\$
C Inertia (6.71 / 2.7% / vol2.8%)	5.851	14.7%	-11.1%	+2.63 T\$

3.2 Pace sensitivity (base capital assumptions)

Scenario	USD/CNY end	Low-end 5y	Net benefit
No appreciation (0%)	7.171	+0.1%	+2.58 T\$
Slow 3%	6.172	-12.7%	+2.63 T\$
Base 6%	5.312	-28.1%	+2.68 T\$
Fast 10%	4.349	-30.0%	+2.75 T\$

3.3 Capital-inflow sensitivity (6% appreciation)

Capital regime	Net benefit (5y)
Conservative (6.5%/yr, weak drain)	+1.99 T\$

Capital regime	Net benefit (5y)
Base (11.1%/yr)	+2.68 T\$
Aggressive (15%/yr, strong drain)	+3.91 T\$

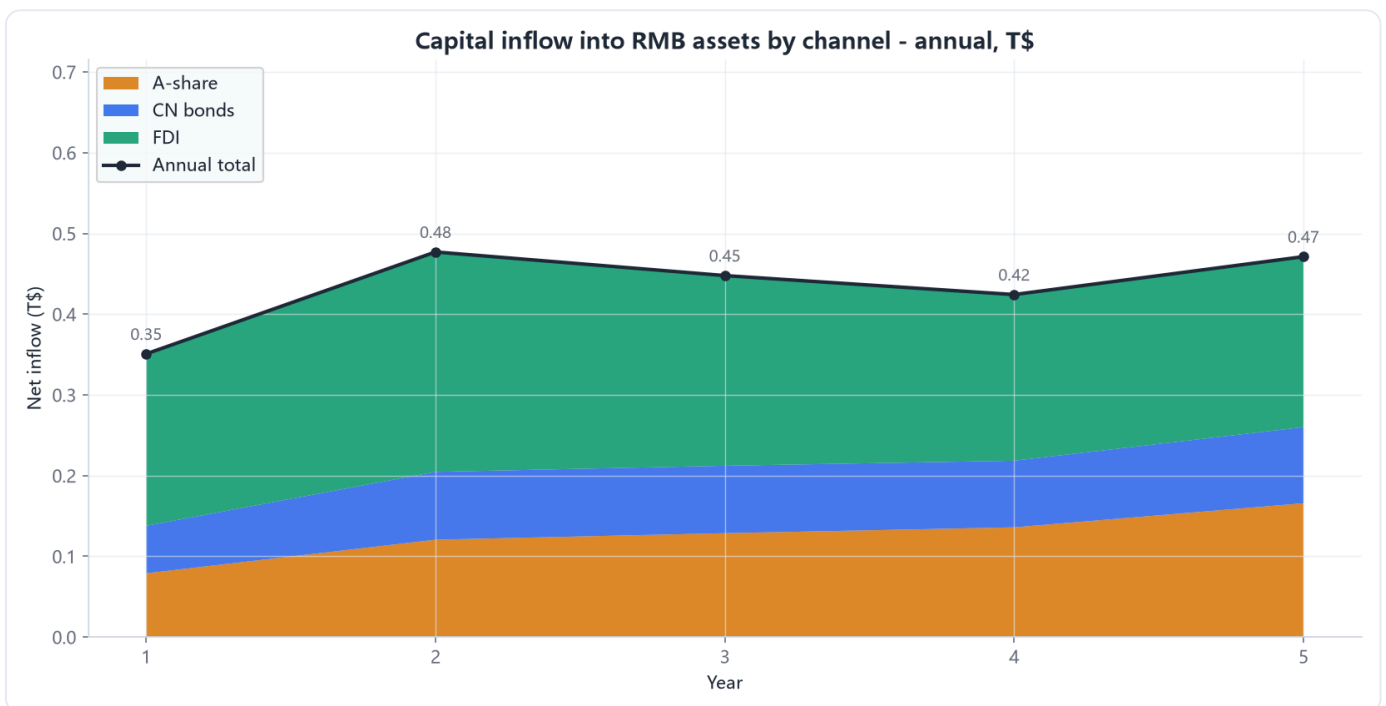
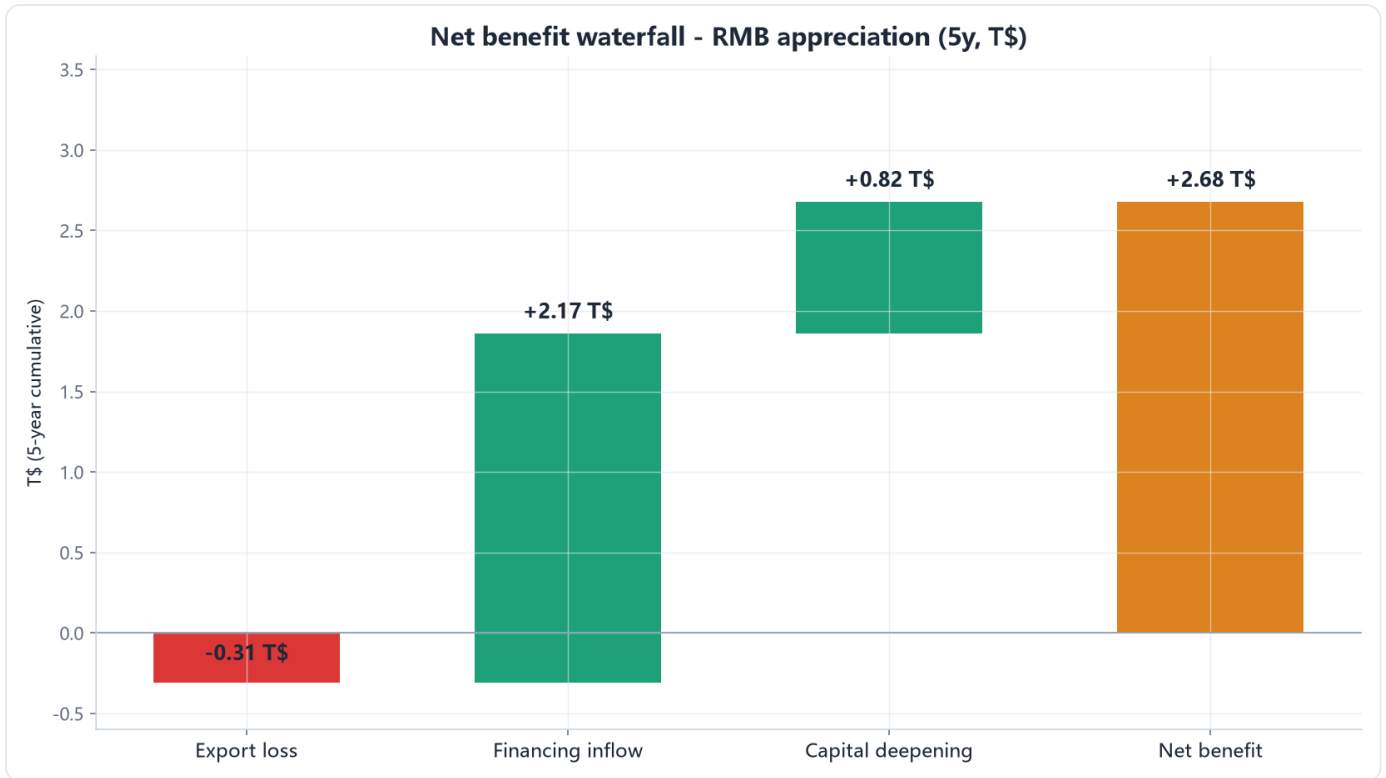
Inflow strength is the largest amplifier of net benefit - and the policy lever: keep the market open and expectations managed so that capital wants to come, dares to stay.

4. Industry profit reallocation (6% base, 5y cumulative)

Sector	5y profit shock
Low-end mfg	-28.1%
High-tech exports	-11.0%
New energy	-3.7%
Consumer	+5.8%
Infrastructure	+3.4%
Property	+7.0%
Banks/Insurance	+3.4%
Semiconductor	+8.7%
Compute/Power	+9.9%
Gold	+13.5%
Metals	+15.2%

Structural read: low-end manufacturing -28.1% (no pricing power); high-tech exports -11.0% (partial pricing power); semiconductor +8.7% and compute/power +9.9% gain from cheaper imported equipment plus capital deepening; metals +15.2% and gold +13.5% gain from RMB purchasing power and global hard-asset pricing; property +7.0% benefits from balance-sheet repair as low-cost long capital replaces short expensive debt. Appreciation is a profit reallocation between old and new economy - losers need industrial-policy buffers, winners get global capital endorsement.

5. Capital inflow - the under-appreciated revenue engine



5.1 Six positive spillover channels

1. **Lower funding costs** - foreign money flattens the long end for private and innovation credit;
2. **Technology spillovers via FDI** - upgrading the high-end supply chain;
3. **Capital-market deepening** - internationalized A-shares and bonds finance new-quality productivity;

4. **Wealth effects** - revaluation repairs household and corporate balance sheets;
5. **RMB internationalization** - settlement and reserve demand, financial autonomy;
6. **Debt-structure optimization** - cheap long capital replaces expensive short debt.

The loop: US fiscal trap makes dollar liquidity scarce - global capital searches for a container - RMB assets (real economy, cheap, policy-controllable) become the preferred one - revaluation plus capital deepening - cheap long capital flows back to the real economy - upgrading, not deindustrialization.

6. Asset implications (5y, median scenario)

Asset	0%	6%	10%	Real-cal. 6%
A-shares (annualized)	+8.0%	+8.2%	+8.7%	+8.2%
CN 10Y bonds	-1.5%	-0.7%	-0.7%	-0.7%
Gold (CNY terms)	+2.9%	-5.0%	-5.0%	-5.0%
Industrial metals	+3.4%	+2.0%	+2.0%	+2.0%

A-shares compound at about +8% across all appreciation speeds (new-quality productivity earnings plus foreign re-rating). CNY-denominated gold is dragged by the exchange rate (USD gold still benefits from de-dollarization and central bank buying). Bonds suffer modest capital losses (1.7% coupon as the floor). **Portfolio read: equity (productivity + resources) beats rates; RMB assets rise as the cleanest "non-dollar duration".**

7. Policy and investment implications

- **Pace:** the net benefit is not very sensitive to 3-10% appreciation - what matters is whether capital flows in. Policy should focus on expectation management, openness (northbound/bond access), and industrial buffers for the most exposed sectors (low-end manufacturing -28.1%).
- **Sectors:** winners are semiconductors/compute, metals and gold (hard assets), banks (spread repair); losers are low-end manufacturing and part of high-tech exports - FX hedging tools and conversion support needed.
- **Allocation:** A-shares (productivity + resources) over metals over bonds; gold best held in USD terms during RMB appreciation. The 48%-cash posture from the GOR framework is a freeze-window defense, not a rejection of this revaluation thesis - survive first, revalue second.

Risks and limitations

- Inflow intensity (conservative 6.5% to aggressive 15% per year) is the largest uncertainty and is assumed, not measured post-2024;
- Northbound daily data ended at the 2024-08 disclosure reform - a dark period, extrapolated with scenario judgment;
- Industry shocks are elasticity approximations without hedging or conversion behavior;
- Export elasticity tested on value (insignificant); volume channel awaits quantity data;
- In a dollar-liquidity crisis, gold and A-shares could fall together; the model does not endogenize freeze-phase path dependence - combine with the GOR framework freeze playbook.